Intro to DBMS & Relational models

Agenda

1. What is a database
2. What is DBMS
3. Types of Databases
   - → Relational DB
   - → Non Relational DB
4. Relational DB
5. Keys:
   - → Super key
   - → Candidate key
   - → Primary key
   - → Composite keys

   NEXT class

6. Foreign key (diff kind of key)

SQL

- → Create a table (CREATE TABLE)
- → add keys to a table (ALTER)

- → How to setup your machine
  - → MySQL 8
  - → MySQL Work bench
  - → Sakila DB

} Needed for following classes

What is a Database

How do you store info about something/someone in daily life?

↳ Dairy / doc → How to search?

↓

rather keep in excel/sheets

(for structured data)

We store what?

→ Expenses

→ finances

→ To do list

→ Groceries

Just like humans

An org also has a lot of data to store so to fetch later

Scaler

student → name, email, grad-year, psp, attendance, resume.

Contests → question, leaderboard, score

batch → students, st-date, cur-module, instructor

We store this preferably in excel.
or file

→ create one file per type of item that I want to store details about.

→ one line per item

students.txt

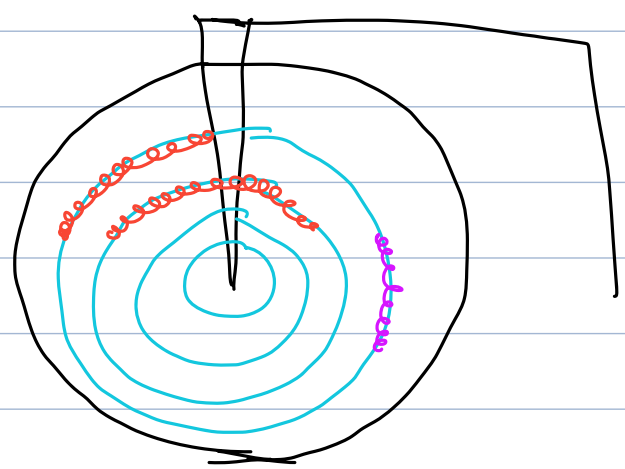
	name	batch	psp	attendance	rank
1.					
2.	Ujj	1	80	75	2
3.	Ravi	2	75	22	5
4.	Shivi	1	88	100	3

We can add new students / rows to this table.

But we also have to retrieve info.

How?

Files are stored in HDD.



This pointer needs to move to read data.

→ You can only read a file sequentially (serially), you cant skip to mid part.

Q Find avg marks of all Abhisheks that I have in my data

→ Go line by line

→ Read one line

→ Split

→ check if name is Abhishek

CONS

① Inefficient → Reading data is very very slow
→ O(N)

② Security → Data is readable by anyone
→ No access control

③ Data Integrity
→ Wrong data types
→ Wrong formatting
→ (also name saved incorrectly like Ujj, wal)

④ You can only read one data at one time.
Concurrency

What is a Database.

Airbase → Plane parking

Military Base → Army / Military stays

Data base ⇒ Data stays

→ where data is stored

Database : Collection of related data

→ Data about items related to each other based on business use case.

Scaler DB → eg. Student, batch, instructor, mentors

↓

for batch id 1, we will have data stored in a separate file

students.txt ⟶ batches.txt

What is DBMS

Software that allows efficient ways to store and retrieve data

along with other necessary features as:

1. Data integrity
2. Concurrency
3. Security etc.

Q find all students with psp b/w 50 & 90

→ DBMS uses adv data structures to store data internally (making queries faster)

Types of Databases (DBMS)

In school how was our data stored?

Notebooks / registers

having tables in them

Attendance Register

name	01/06	02/06	03/06	04/06	05/06	.	.
Rohit							

Tables are very natural / simplest way to save data.

DBMS

Relational database

→ Collection of tables related to each other

each col is attr of that item

Students

name	age	psp	batch

→ diff item in each row

batches

id	name	st. date

Non-relational database [NoSQL DB]

→ Collection of related data stored in form OTHER THAN tables.

HLD
- → graph database
- → columnar
- → document
- → key-value pairs

[Break]

Properties of relational dB

① Rel^n dB represent data as collection of tables with each table storing info about something.

eg. Students, instructor, batches

② Every row is unique. In any table no 2 rows should have the same values for all col". Else difficult to identify someone uniquely.

students

name	age
Alok	20
Ujjwal	26
Ujjwal	26

? who is who

③ All of values in a col should be of same data type.

④ Values in cells are atomic.
↳ Smallest indivisible unit.

SQL Standards → cant have arrays, maps etc in a rel" collection.

(Lately SQL dBs allow non-atomic values in them)
→ even SQL DB that allows non-atomic values are typically not efficient

⑤ The column seq in queries is not guaranteed.

Student

roll no.	Name	email	Phone

select * from students
where roll no = 33;

[we cant assume which the first or last col will be]

Many SQL dB return col in the order in which they were present in the table.

Never assume order in col ⇒ always specify the order you want.

⑥ The order of rows is also not guaranteed.

1	Ujjwal
2	Amit

next query →

1	Amit
2	Ujjwal

→ If you need rows in a particular order specify the order.

→ In reality many SQL dB guarantee that rows will be ordered by Primary Keys.
[But dont assume it].

⑦ No 2 col can have same name.

Keys

→ every row should be unique.

attendance students

roll no	name	year_of_birth	01/06	02/06	03/06
1	Abhishek	1997	N	Y	Y
2	Abhishek	1997	N	Y	Y

How to identify Abhishek ?

Key → A col or a set of col of a table whose values are guaranteed to be able to identify a row uniquely.

year_of_birth ✗

name ✗

roll_no ✓

Super key

A col or a set of col whose value is guaranteed to uniquely identify an item

in a table.

Student

name	psp	email	batch	phone_number

(phone No) ✓
(email) ✓
(name) ✗
(psp) ✗
(batch) ✗
(phone No, email) ✓
(phone No, name) ✓
(email, name) ✓
(phone No, psp) ✓
(phone No, psp, batch) ✓
(psp, batch) ✗
(psp, batch, name) ✗

super keys